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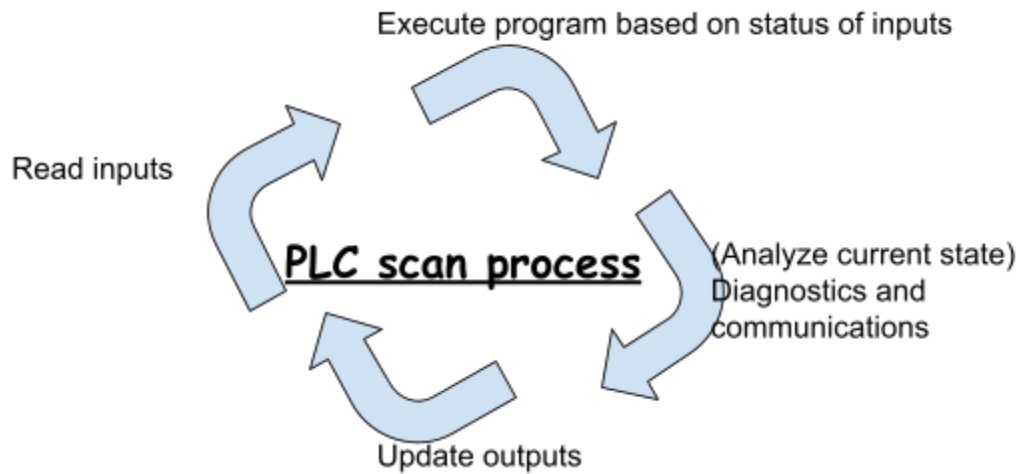
Lab 7

3/27/25

PLC Implementation of the Motor Starter H-Bridge

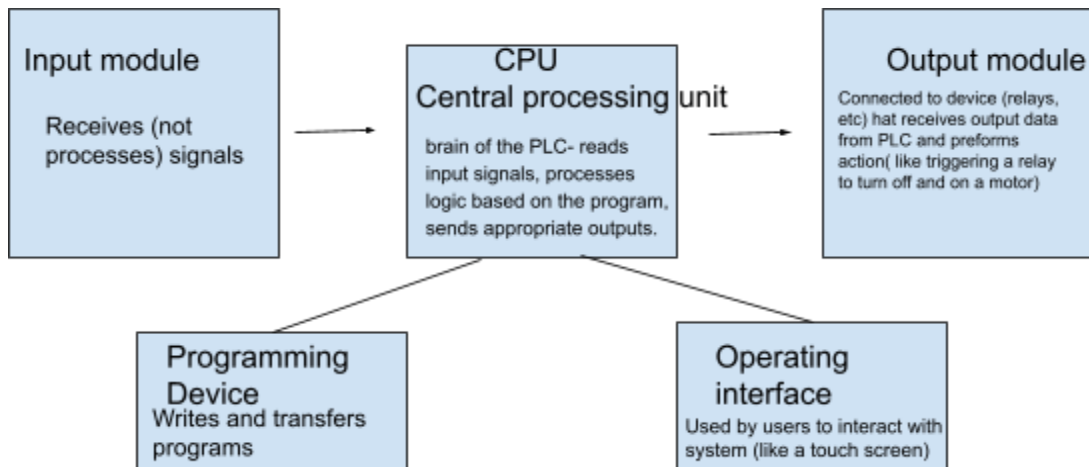
Introduction

This lab focused on recreating a Forward-Reverse Motor Starter using a PLC and integrating input and output devices. PLCs (programmable logic controllers) are like mini computers that are used to control machines and processes. The PLC first gets information (inputs), then makes decisions (the program) and then it sends out commands (outputs). We use PLCs for many reasons- they are very customizable (because to change the PLC function you just update the program), they react quickly to changes (this is good for safety) and they are good for automation (helping things run automatically with little human interference and no need for people to do stuff like press buttons) and are reliable. PLC's essentially replace human input and action into industrial electronics, which is a great safety feature, saves money and is convenient.



Overall cycle time depends on size of the program, number of I/Os (inputs and outputs) and the amount of communication required

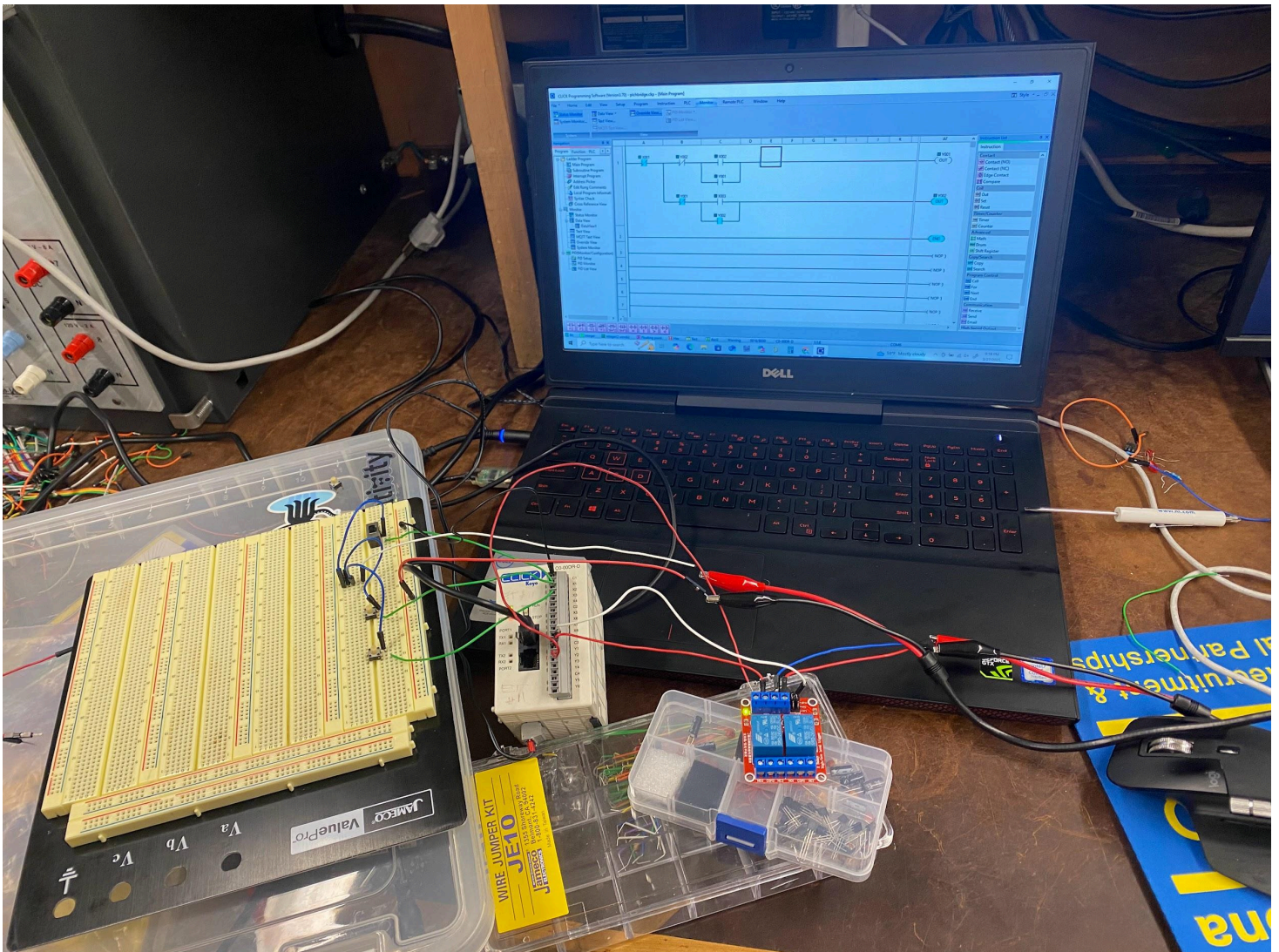
IMG 1. An explanation of the PLC process.



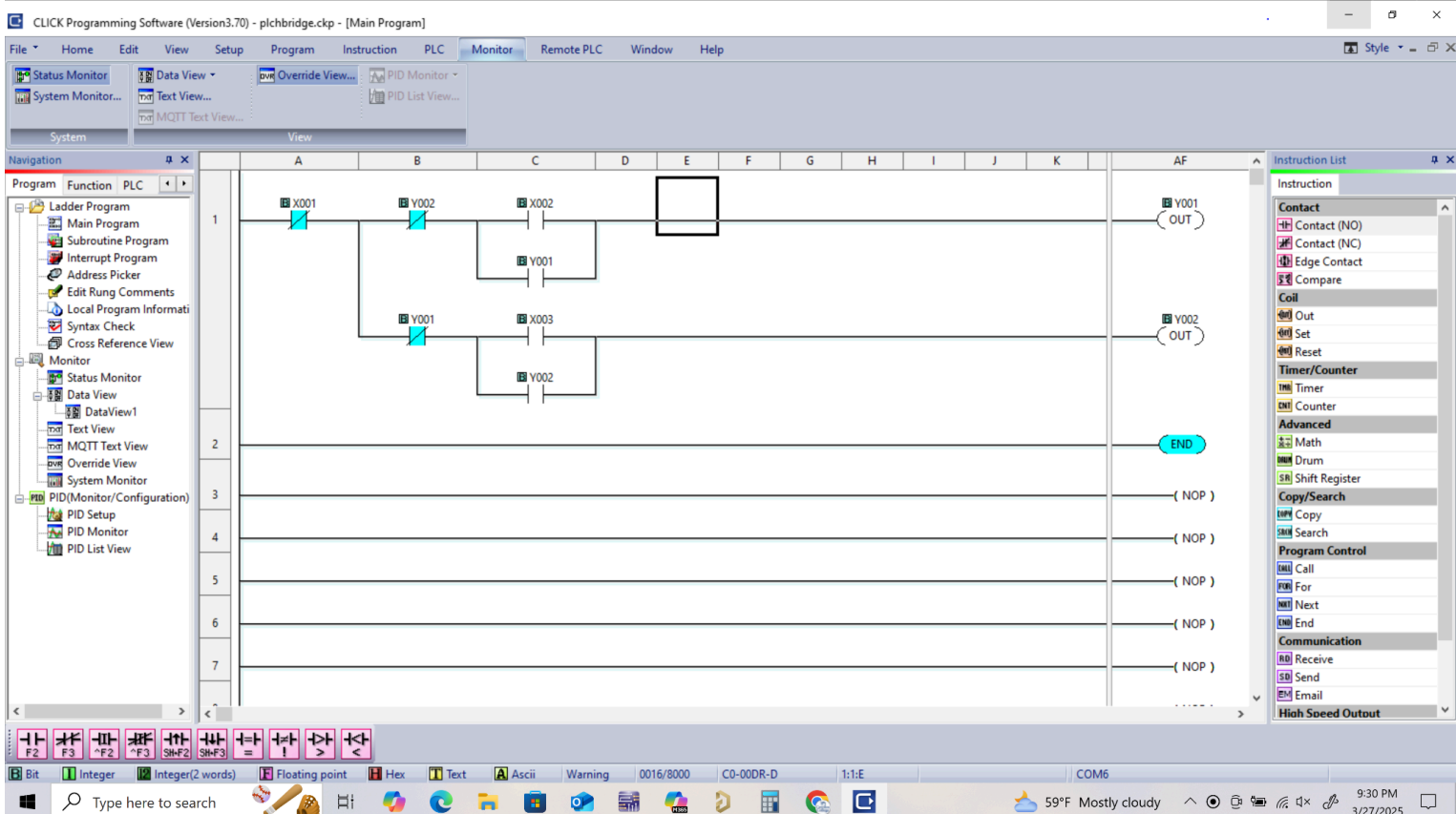
IMG 2. Aspects of the PLC.

Part 1:

Evidence of build:



IMG 3. Physical circuit



IMG 4. Schematic of A and C relays on PLC, programmed on CLICK programming software

Video link to functioning circuit:

<https://drive.google.com/file/d/12c9GJNoupYFqcOdTy5dmRtc4KoF-fPdN/view?usp=sharing>

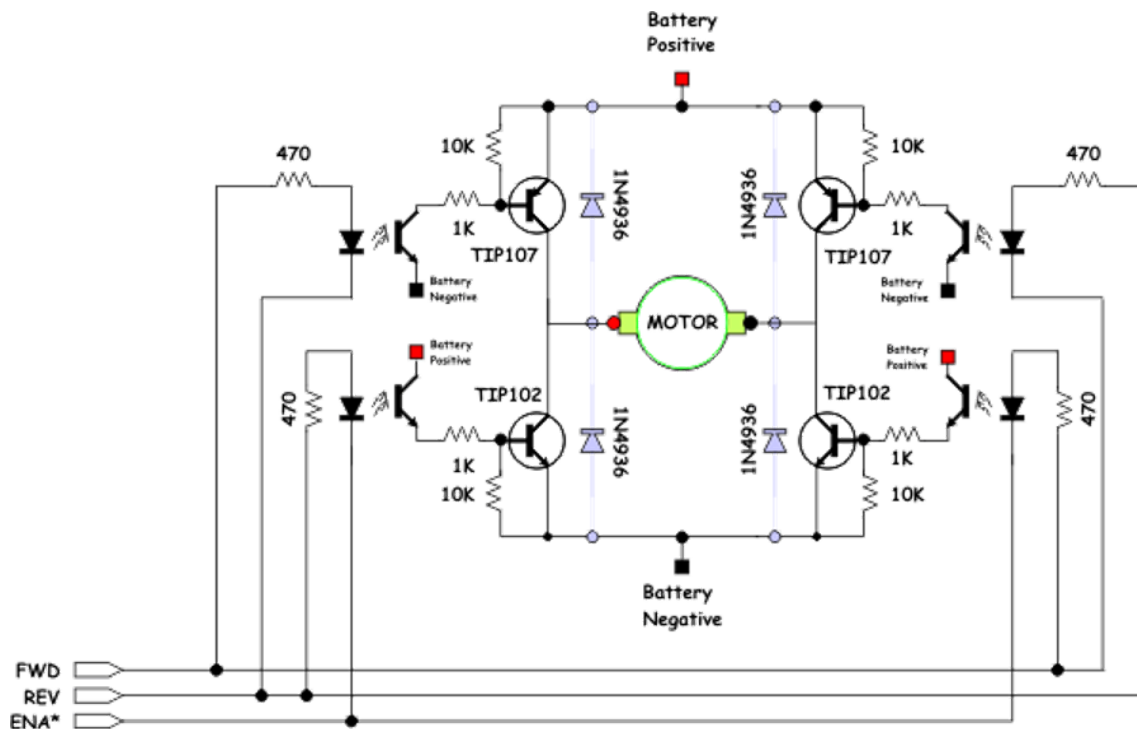
The PLC required three switches: a stop switch (normally closed, to stop the whole circuit in case of an emergency), a forward switch (normally open), and a reverse switch (normally open), all single pole single throw switches. On the relay, X1, X2, and X3 went to the switches, which controlled the PLC and Arduino relays. Instead of only using physical relay components for relays A, B, C, and D, like the circuit from previous labs, the PLC carried out the programmed logic of relays A and C. Relays B and D were relays from Arduino. Instead of only using physical relay components for relay A, B, C and D like the circuit from previous labs, the PLC carried out the programmed logic of relays A and C.

Relays B and D were relays from Arduino, and high and low (normally open and normally closed) was switched on the Arduino relays.. The PLC software handled the interpretation of the inputs going to relay A and C, allowing them to function as normally open (NO) or normally closed (NC) operations of the internal switches of the relay. The PLC and relays B and D have their power, and the breadboard has its power. Although no motor was connected in this circuit, the relay switching is correct and would control the motor if attached.

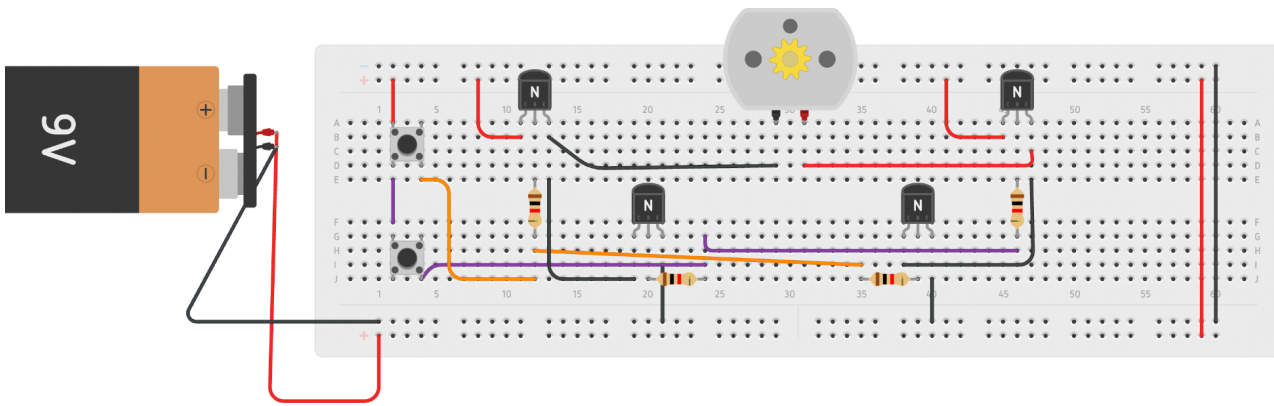
Part 2: Analyzing an Electronic H-Bridge using Transistors

We referenced the H-bridge article provided by Chuck McManis at

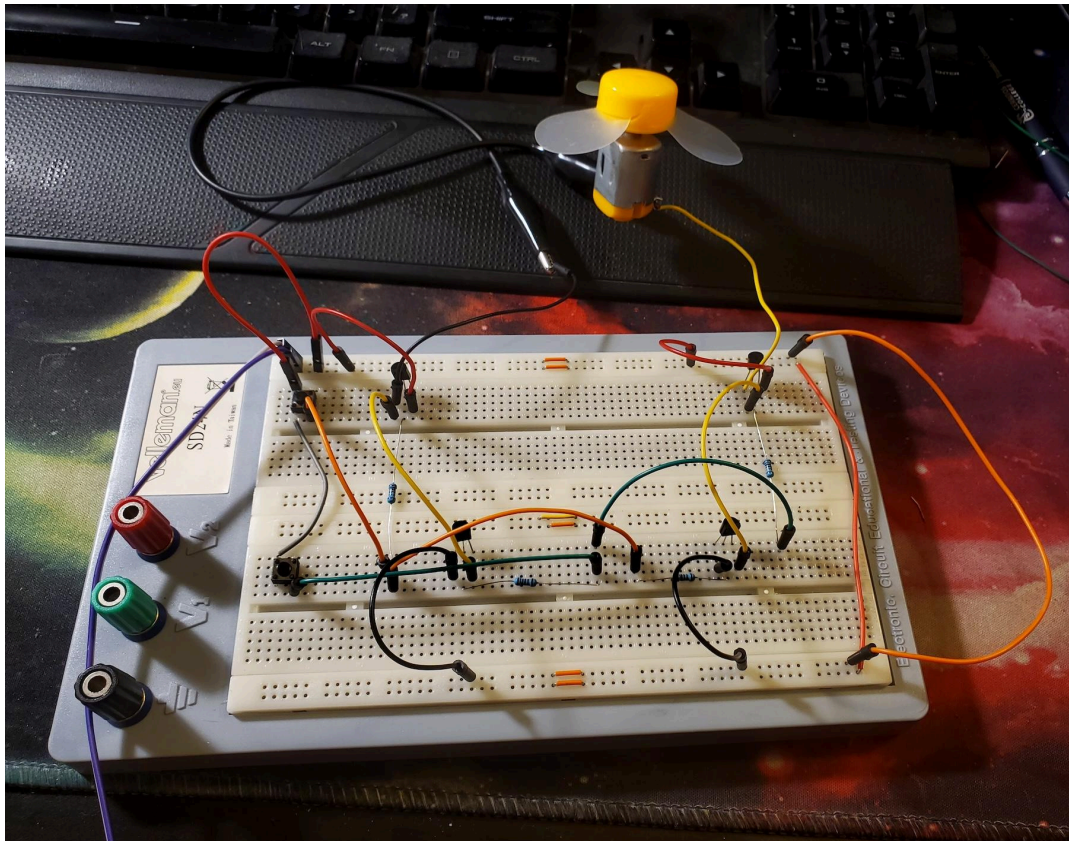
<http://www.mcmanis.com/chuck/robotics/tutorial/h-bridge/bjt-circuit.html>



IMG 5. Schematic of H-bridge from Chuck McManis



IMG 6. *From simulation of H-bridge.*



IMG 7. *Physical H-bridge circuit*

Video link to functioning circuit:

<https://drive.google.com/file/d/1VhysPxn4PPRgO8KuPcBx2rqysAg4BUkV/view?usp=sharing>

Conclusion

This lab showed us firsthand the usefulness and ease of programming and working with PLC's. We experienced the customization aspects and the ease of fixing the program to alter the circuit, and it was clear to see why using PLC for industrial electronics improves convenience and safety. With the exception of updating and maintaining the PLC's, using PLCs removes quite a bit of human interaction with the electronics. A key challenge in this lab was correctly wiring all components, particularly ensuring proper standard connections. Initially, we did not correctly connect components to the common, and additional difficulties arose in configuring the Click PLC program. We also had trouble orienting the switches and understanding the schematic. Despite these challenges, the final setup successfully demonstrated relay switching that would control a motor if one were connected.